Validate Subsequence

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The problem

- Is `sequence` obtainable from `array` by deleting elements without reordering?

Intuition

- A subsequence keeps order, so we just try to match the sequence's elements one by one while scanning the array.

Approach

- 1. Keep an index into the sequence
- 2. Scan the array
- 3. Advance the index on every match
- 4. Valid if the index reaches the end

Java solution

```java
public class ValidateSubsequence {
    public static boolean isValidSubsequence(List<Integer> array, List<Integer> sequence) {
        int j = 0;
        for (int i = 0; i < array.size() && j < sequence.size(); i++)
            if (array.get(i).equals(sequence.get(j))) j++;
        return j == sequence.size();
    }
}
```

Complexity

- Time: O(n)
- Space: O(1)

Recap

- A subsequence keeps order, so we just try to match the sequence's elements one by one while scanning the array.
- Complexity: O(n)